MYSQL ASSIGNMENT 2

CLAUSES AND JOINS

PREPARED BY:

AMRA ABDUL HAMEED

The Database employee is used here, from the previous assignment. The table Employee Data is populated using the given DML commands.

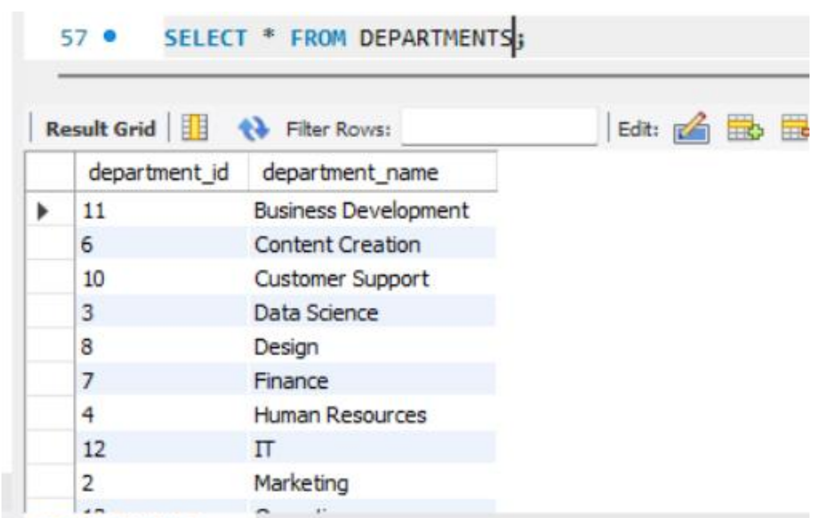
```
USE employee;

INSERT INTO departments (department_id, department_name) VALUES
(1, 'Software Development'),
(2, 'Marketing'),
(3, 'Data Science'),
(4, 'Human Resources'),
(5, 'Product Management'),
(6, 'Content Creation'),
(7, 'Finance'),
(8, 'Design'),
(9, 'Research and Development'),
(10, 'Customer Support'),
(11, 'Business Development'),
(12, 'IT'),
(13, 'Operations');

INSERT INTO location (location) VALUES
('Chennai'),
('Bangalore'),
('Hyderabad'),
('Pune');

INSERT INTO employees (employee_id, employee_name, gender, age, hire_date,
designation, department_id, location_id, salary) VALUES
(5001, 'Vihaan Singh', 'M', 27, '2015-01-20', 'Data Analyst', 3, 4,
60000),
(5002, 'Reyansh Singh', 'M', 31, '2015-03-10', 'Network Engineer', 12, 1,
80000),
(5003, 'Aaradhya Iyer', 'F', 26, '2015-05-20', 'Customer Support
Executive', 10, 2, 45000),
(5004, 'Kiara Malhotra', 'F', 29, '2015-07-05', NULL, 8, 3, 70000),
(5005, 'Anvi Chaudhary', 'F', 25, '2015-09-11', 'Business Development
Executive', 11, 1, 55000),
(5006, 'Dhruv Shetty', 'M', 28, '2015-11-20', 'UI Developer', 8, 2,
65000),
(5007, 'Anushka Singh', 'F', 32, '2016-01-15', 'Marketing Manager', 2, 3,
90000),
(5008, 'Diya Jha', 'F', 27, '2016-03-05', 'Graphic Designer', 8, 4,
70000),
(5009, 'Kiaan Desai', 'M', 30, '2016-05-20', 'Sales Executive', 11, 3,
55000),
(5010, 'Atharv Yadav', 'M', 29, '2016-07-10', 'Systems Administrator', 12,
4, 80000),
(5011, 'Saanvi Patel', 'F', 28, '2016-09-20', 'Marketing Analyst', 2, 1,
60000),
(5012, 'Myra Verma', 'F', 26, '2016-11-05', 'Operations Manager', 13, 2,
95000),
(5013, 'Arnav Rao', 'M', 33, '2017-01-20', 'Customer Success Manager', 10,
3, 75000),
```

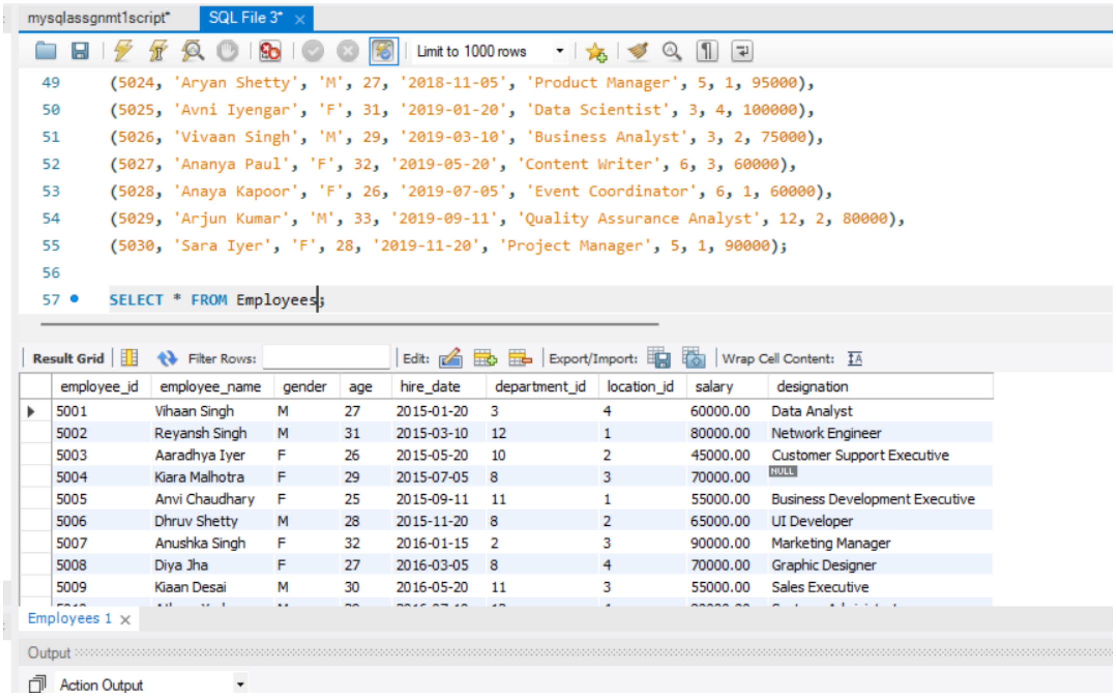
```
(5014, 'Vihaan Mohan', 'M', 30, '2017-03-10', 'Supply Chain Analyst', 10,
2, 60000),
(5015, 'Ishaan Kumar', 'M', 27, '2017-05-20', 'Financial Analyst', 7, 1,
85000),
(5016, 'Zoya Khan', 'F', 31, '2017-07-05', 'Legal Counsel', 4, 4, 100000),
(5017, 'Kabir Nair', 'M', 28, '2017-09-11', 'IT Support Specialist', 12,
2, 80000),
(5018, 'Ishan Mishra', 'M', 25, '2017-11-20', 'Research Scientist', 9, 3,
75000),
(5019, 'Ishika Patel', 'F', 29, '2018-01-15', 'Talent Acquisition
Specialist', 4, 4, 55000),
(5020, 'Aarav Nair', 'M', 32, '2018-03-05', 'Software Engineer', 1, 1,
90000),
(5021, 'Advik Kapoor', 'M', 26, '2018-05-20', 'Finance Analyst', 7, 3,
85000),
(5022, 'Aadhya Iyengar', 'F', 28, '2018-07-10', 'HR Specialist', 4, 4,
60000),
(5023, 'Anika Paul', 'F', 30, '2018-09-20', 'Public Relations Specialist',
2, 2, 70000),
(5024, 'Aryan Shetty', 'M', 27, '2018-11-05', 'Product Manager', 5, 1,
95000),
(5025, 'Avni Iyengar', 'F', 31, '2019-01-20', 'Data Scientist', 3, 4,
100000),
(5026, 'Vivaan Singh', 'M', 29, '2019-03-10', 'Business Analyst', 3, 2,
75000),
(5027, 'Ananya Paul', 'F', 32, '2019-05-20', 'Content Writer', 6, 3,
60000),
(5028, 'Anaya Kapoor', 'F', 26, '2019-07-05', 'Event Coordinator', 6, 1,
60000),
(5029, 'Arjun Kumar', 'M', 33, '2019-09-11', 'Quality Assurance Analyst',
12, 2, 80000),
(5030, 'Sara Iyer', 'F', 28, '2019-11-20', 'Project Manager', 5, 1,
90000);
```



```
57 • SELECT * FROM DEPARTMENTS;
```

Result Grid | Filter Rows: | Edit:

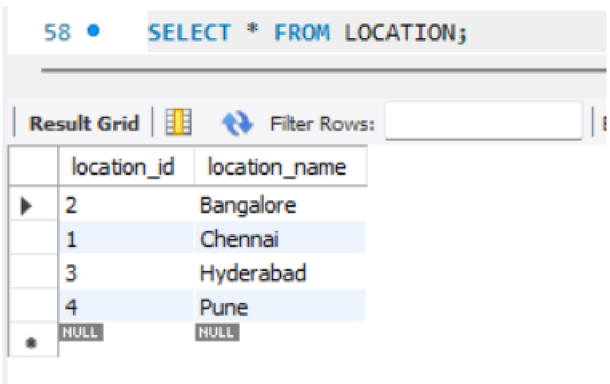
department_id	department_name
11	Business Development
6	Content Creation
10	Customer Support
3	Data Science
8	Design
7	Finance
4	Human Resources
12	IT
2	Marketing



```
(5024, 'Aryan Shetty', 'M', 27, '2018-11-05', 'Product Manager', 5, 1, 95000),
(5025, 'Avni Iyengar', 'F', 31, '2019-01-20', 'Data Scientist', 3, 4, 100000),
(5026, 'Vivaan Singh', 'M', 29, '2019-03-10', 'Business Analyst', 3, 2, 75000),
(5027, 'Ananya Paul', 'F', 32, '2019-05-20', 'Content Writer', 6, 3, 60000),
(5028, 'Anaya Kapoor', 'F', 26, '2019-07-05', 'Event Coordinator', 6, 1, 60000),
(5029, 'Arjun Kumar', 'M', 33, '2019-09-11', 'Quality Assurance Analyst', 12, 2, 80000),
(5030, 'Sara Iyer', 'F', 28, '2019-11-20', 'Project Manager', 5, 1, 90000);

SELECT * FROM Employees;
```

employee_id	employee_name	gender	age	hire_date	department_id	location_id	salary	designation
5001	Vihaan Singh	M	27	2015-01-20	3	4	60000.00	Data Analyst
5002	Reyansh Singh	M	31	2015-03-10	12	1	80000.00	Network Engineer
5003	Aaradhya Iyer	F	26	2015-05-20	10	2	45000.00	Customer Support Executive
5004	Kiara Malhotra	F	29	2015-07-05	8	3	70000.00	NULL
5005	Anvi Chaudhary	F	25	2015-09-11	11	1	55000.00	Business Development Executive
5006	Dhruv Shetty	M	28	2015-11-20	8	2	65000.00	UI Developer
5007	Anushka Singh	F	32	2016-01-15	2	3	90000.00	Marketing Manager
5008	Diya Jha	F	27	2016-03-05	8	4	70000.00	Graphic Designer
5009	Kiaan Desai	M	30	2016-05-20	11	3	55000.00	Sales Executive



```
SELECT * FROM LOCATION;
```

location_id	location_name
2	Bangalore
1	Chennai
3	Hyderabad
4	Pune
NULL	NULL

Execute the queries in this assignment after inserting the data.

1. Distinct Values:

◉ a query to retrieve distinct salaries from the Employees table.

SELECT DISTINCT salary FROM employees;

```sql
SELECT DISTINCT salary
FROM employees;
```

Result Grid | Filter Rows:

salary
60000.00
80000.00
45000.00
70000.00
55000.00
65000.00
90000.00
95000.00
75000.00

2. Alias (AS):

◉ Provide aliases for the "age" and "salary" columns as "Employee_Age" and "Employee_Salary", respectively.

SELECT

  age AS Employee_Age,

  salary AS Employee_Salary

FROM employees;

```sql
SELECT
    age AS Employee_Age,
    salary AS Employee_Salary
FROM employees;
```

sult Grid | Filter Rows: | Export:

Employee_Age	Employee_Salary
27	60000.00
31	80000.00
26	45000.00
29	70000.00
25	55000.00
28	65000.00
32	90000.00
27	70000.00
30	55000.00

loyees 5 ×

3. Where Clause & Operators:

◉ Retrieve employees with a salary greater than ₹50000 and hired before 2016-01-01.

SELECT *

FROM employees

WHERE salary > 50000

AND hire_date < '2016-01-01';

```
SELECT *
FROM employees
WHERE salary > 50000
  AND hire_date < '2016-01-01';
```

Result Grid | Filter Rows: | Edit: | Export/Import: | Wrap Cell Content:

employee_id	employee_name	gender	age	hire_date	department_id	location_id	salary	designation
5001	Vihaan Singh	M	27	2015-01-20	3	4	60000.00	Data Analyst
5002	Reyansh Singh	M	31	2015-03-10	12	1	80000.00	Network Engineer
5004	Kiara Malhotra	F	29	2015-07-05	8	3	70000.00	NULL
5005	Anvi Chaudhary	F	25	2015-09-11	11	1	55000.00	Business Development Executive
5006	Dhruv Shetty	M	28	2015-11-20	8	2	65000.00	UI Developer
NULL	NULL	NULL	NULL	NULL	NULL	NULL	NULL	NULL

◉ Find the employee whose designation is missing and fill it with "Data Scientist".

```
SELECT *
FROM employees
WHERE designation IS NULL;
```

Result Grid | Filter Rows: | Edit: | Export/Import: | Wrap Cell Content:

employee_id	employee_name	gender	age	hire_date	department_id	location_id	salary	designation
5004	Kiara Malhotra	F	29	2015-07-05	8	3	70000.00	NULL
NULL	NULL	NULL	NULL	NULL	NULL	NULL	NULL	NULL

```
UPDATE employees
SET designation = 'Data Scientist'
WHERE employee_id = 5004;
```

```sql
SELECT *
FROM employees
WHERE employee_id = 5004;
```

Result Grid | Filter Rows: | Edit: | Export/Import: | Wrap Cell Content:

employee_id	employee_name	gender	age	hire_date	department_id	location_id	salary	designation
5004	Kiara Malhotra	F	29	2015-07-05	8	3	70000.00	Data Scientist
NULL	NULL	NULL	NULL	NULL	NULL	NULL	NULL	NULL

Sorting and Grouping Data:

1. ORDER BY:

◉ Find employees sorted by department ID in ascending order and salary in descending order.

```sql
SELECT *
FROM employees
ORDER BY department_id ASC, salary DESC;
```

Result Grid | Filter Rows: | Edit: | Export/Import: | Wrap Cell Content:

employee_id	employee_name	gender	age	hire_date	department_id	location_id	salary	designation
5020	Aarav Nair	M	32	2018-03-05	1	1	90000.00	Software Engineer
5007	Anushka Singh	F	32	2016-01-15	2	3	90000.00	Marketing Manager
5023	Anika Paul	F	30	2018-09-20	2	2	70000.00	Public Relations Specialist
5011	Saanvi Patel	F	28	2016-09-20	2	1	60000.00	Marketing Analyst
5025	Avni Iyengar	F	31	2019-01-20	3	4	100000.00	Data Scientist
5026	Vivaan Singh	M	29	2019-03-10	3	2	75000.00	Business Analyst
5001	Vihaan Singh	M	27	2015-01-20	3	4	60000.00	Data Analyst
5016	Zoya Khan	F	31	2017-07-05	4	4	100000.00	Legal Counsel
5022	Aadhya Iyengar	F	28	2018-07-10	4	4	60000.00	HR Specialist

employees 9 ×

Output

2. LIMIT:

◉ Display the first 5 employees hired in the year 2018.

SELECT *

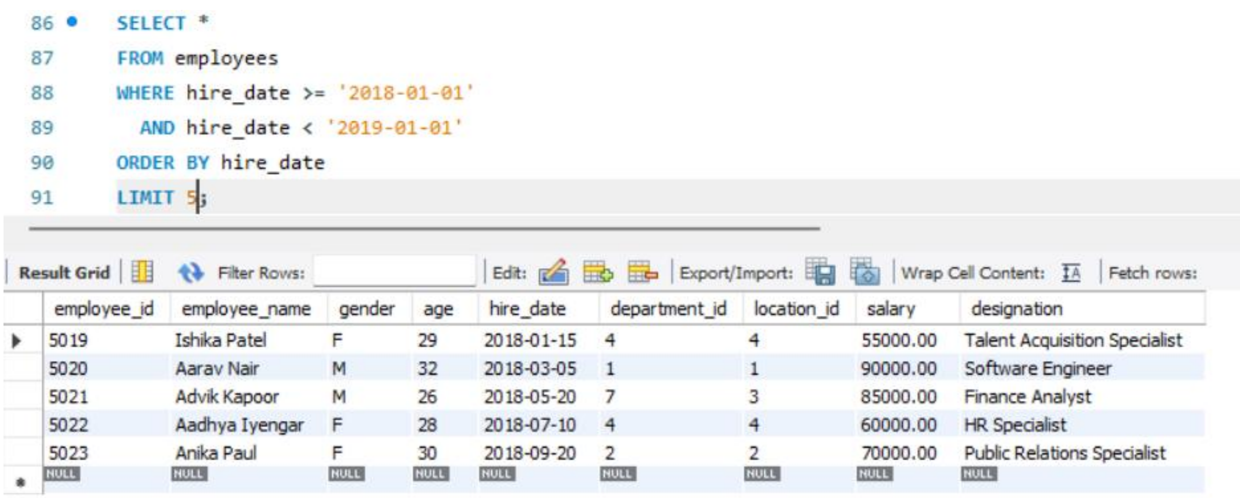
FROM employees

WHERE hire_date >= '2018-01-01'

AND hire_date < '2019-01-01'

ORDER BY hire_date

LIMIT 5;



```sql
SELECT *
FROM employees
WHERE hire_date >= '2018-01-01'
  AND hire_date < '2019-01-01'
ORDER BY hire_date
LIMIT 5;
```

Result Grid | Filter Rows: | Edit: | Export/Import: | Wrap Cell Content: | Fetch rows:

employee_id	employee_name	gender	age	hire_date	department_id	location_id	salary	designation
5019	Ishika Patel	F	29	2018-01-15	4	4	55000.00	Talent Acquisition Specialist
5020	Aarav Nair	M	32	2018-03-05	1	1	90000.00	Software Engineer
5021	Advik Kapoor	M	26	2018-05-20	7	3	85000.00	Finance Analyst
5022	Aadhya Iyengar	F	28	2018-07-10	4	4	60000.00	HR Specialist
5023	Anika Paul	F	30	2018-09-20	2	2	70000.00	Public Relations Specialist
NULL	NULL	NULL	NULL	NULL	NULL	NULL	NULL	NULL

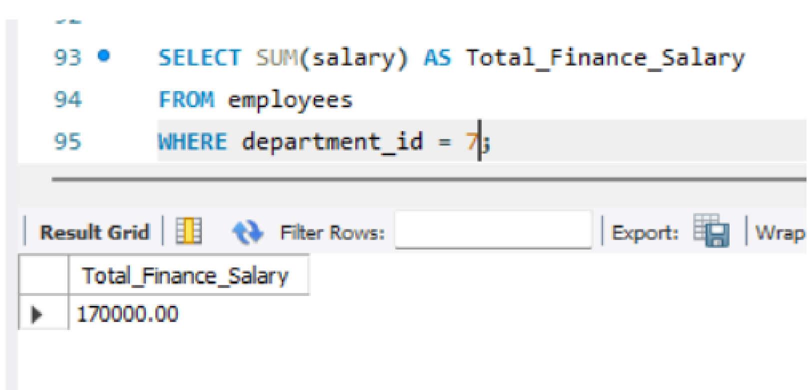
3. Aggregate Functions:

◉ Calculate the sum of all salaries in the Finance department.

SELECT SUM(salary) AS Total_Finance_Salary

FROM employees

WHERE department_id = 7;



```sql
SELECT SUM(salary) AS Total_Finance_Salary
FROM employees
WHERE department_id = 7;
```

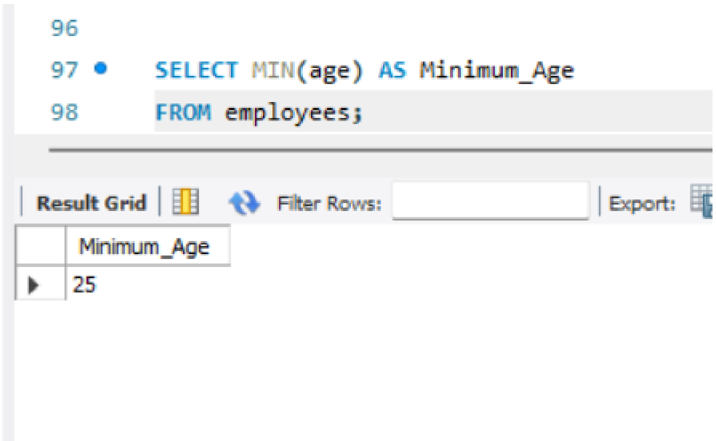
Result Grid | Filter Rows: | Export: | Wrap

Total_Finance_Salary
170000.00

◉ Find the minimum age among all employees.

SELECT MIN(age) AS Minimum_Age

FROM employees;



4. GROUP BY:

◉ List the maximum salary for each location.

SELECT

l.location_name,

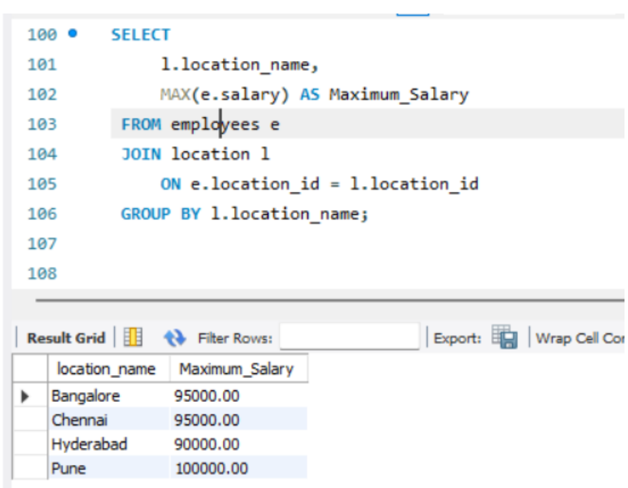
MAX(e.salary) AS Maximum_Salary

FROM employees e

JOIN location l

ON e.location_id = l.location_id

GROUP BY l.location_name;



◉ Calculate the average salary for each designation containing the word 'Analyst'.

SELECT

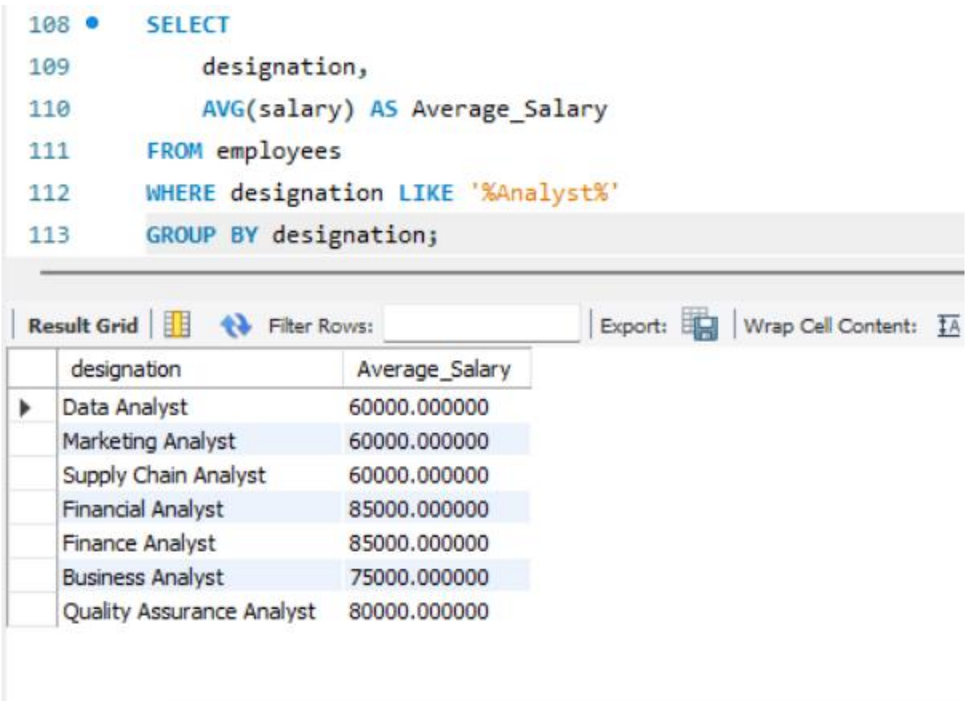
designation,

AVG(salary) AS Average_Salary

FROM employees

WHERE designation LIKE '%Analyst%'

GROUP BY designation;




```
SELECT
    designation,
    AVG(salary) AS Average_Salary
FROM employees
WHERE designation LIKE '%Analyst%'
GROUP BY designation;
```

designation	Average_Salary
Data Analyst	60000.000000
Marketing Analyst	60000.000000
Supply Chain Analyst	60000.000000
Financial Analyst	85000.000000
Finance Analyst	85000.000000
Business Analyst	75000.000000
Quality Assurance Analyst	80000.000000

5. HAVING:

◉ Find departments with less than 3 employees.

SELECT

d.department_id,

d.department_name,

COUNT(e.employee_id) AS Employee_Count

FROM departments d

LEFT JOIN employees e

ON d.department_id = e.department_id

GROUP BY

d.department_id,

d.department_name

HAVING COUNT(e.employee_id) < 3;

```
    d.department_id,
    d.department_name,
    COUNT(e.employee_id) AS Employee_Count
FROM departments d
LEFT JOIN employees e
    ON d.department_id = e.department_id
GROUP BY
    d.department_id,
    d.department_name
HAVING COUNT(e.employee_id) < 3;
```

Result Grid | Filter Rows: | Export: | Wrap Cell Content:

department_id	department_name	Employee_Count
11	Business Development	2
6	Content Creation	2
7	Finance	2
13	Operations	1
5	Product Management	2
9	Research and Development	1
1	Software Development	1

◉ Find locations with female employees whose average age is below 30.

SELECT

l.location_name,

AVG(e.age) AS Average_Female_Age

FROM employees e

JOIN location l

ON e.location_id = l.location_id

WHERE e.gender = 'F'

GROUP BY l.location_name

HAVING AVG(e.age) < 30;

```
SELECT
    l.location_name,
    AVG(e.age) AS Average_Female_Age
FROM employees e
JOIN location l
    ON e.location_id = l.location_id
WHERE e.gender = 'F'
GROUP BY l.location_name
HAVING AVG(e.age) < 30;
```

Result Grid | Filter Rows: | Export: | Wrap

location_name	Average_Female_Age
Bangalore	27.3333
Chennai	26.7500
Pune	29.2000

Joins:

1. Inner Join:

◉ List employee names, their designations, and department names where employees are assigned to a department.

```
SELECT
    e.employee_name,
    e.designation,
    d.department_name
FROM employees e
INNER JOIN departments d
    ON e.department_id = d.department_id;
```

```sql
SELECT
    e.employee_name,
    e.designation,
    d.department_name
FROM employees e
INNER JOIN departments d
    ON e.department_id = d.department_id;
```

Result Grid | Filter Rows: | Export: | Wrap Cell Content:

employee_name	designation	department_name
Anvi Chaudhary	Business Development Executive	Business Development
Kiaan Desai	Sales Executive	Business Development
Ananya Paul	Content Writer	Content Creation
Anaya Kapoor	Event Coordinator	Content Creation
Aaradhya Iyer	Customer Support Executive	Customer Support
Arnav Rao	Customer Success Manager	Customer Support
Vihaan Mohan	Supply Chain Analyst	Customer Support
Vihaan Singh	Data Analyst	Data Science

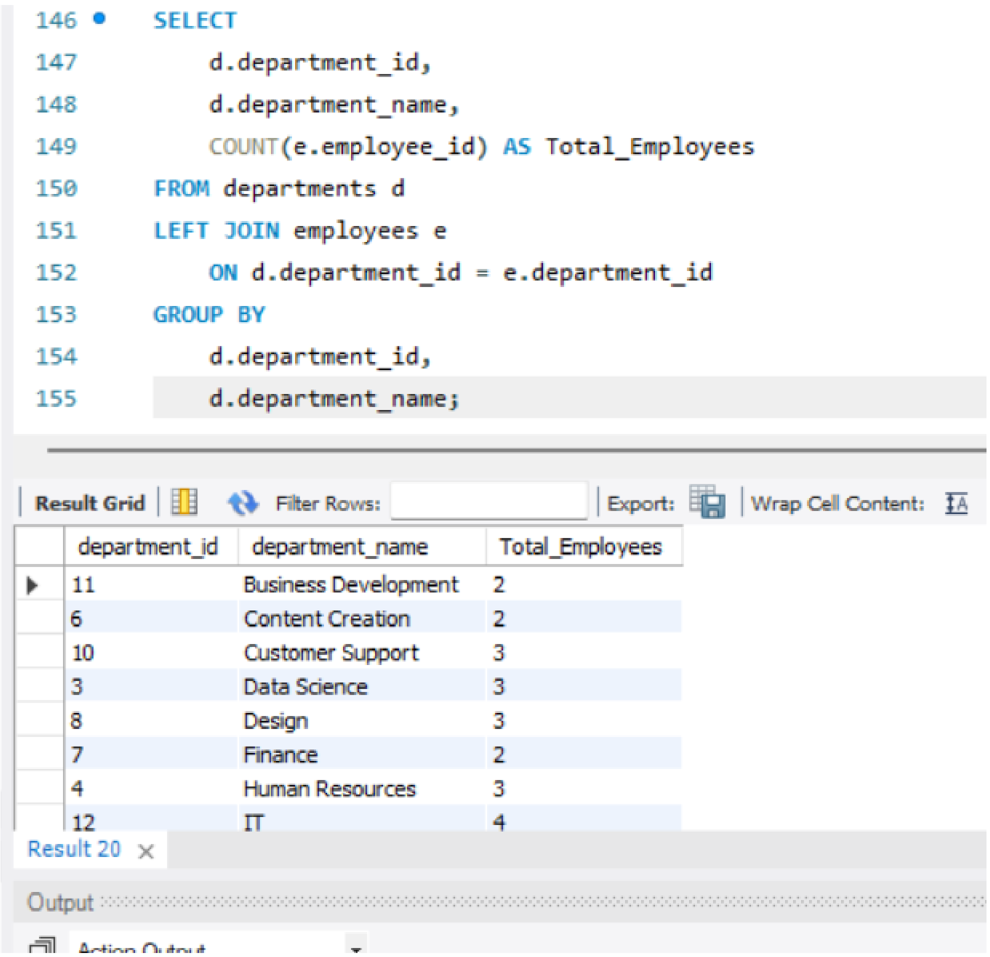
Result 19

Output

2. Left Join:

◉ List all departments along with the total number of employees in each department, including departments with no employees.

```sql
SELECT
    d.department_id,
    d.department_name,
    COUNT(e.employee_id) AS Total_Employees
FROM departments d
LEFT JOIN employees e
    ON d.department_id = e.department_id
GROUP BY
    d.department_id,
    d.department_name;
```



```sql
SELECT
    d.department_id,
    d.department_name,
    COUNT(e.employee_id) AS Total_Employees
FROM departments d
LEFT JOIN employees e
    ON d.department_id = e.department_id
GROUP BY
    d.department_id,
    d.department_name;
```


department_id	department_name	Total_Employees
11	Business Development	2
6	Content Creation	2
10	Customer Support	3
3	Data Science	3
8	Design	3
7	Finance	2
4	Human Resources	3
12	IT	4

3. Right Join:

◉ Display all locations along with the names of employees assigned to each location. If no employees are assigned to a location, display NULL for employee name.

```sql
SELECT
    l.location_name,
    e.employee_name
FROM employees e
RIGHT JOIN location l
    ON e.location_id = l.location_id;
```

```sql
SELECT
    l.location_name,
    e.employee_name
FROM employees e
RIGHT JOIN location l
    ON e.location_id = l.location_id;
```

Result Grid | Filter Rows: | Export: | Wrap Cell Content:

location_name	employee_name
Bangalore	Aaradhya Iyer
Bangalore	Dhruv Shetty
Bangalore	Myra Verma
Bangalore	Vihaan Mohan
Bangalore	Kabir Nair
Bangalore	Anika Paul
Bangalore	Vivaan Singh
Bangalore	Arjun Kumar

Result 21

Output